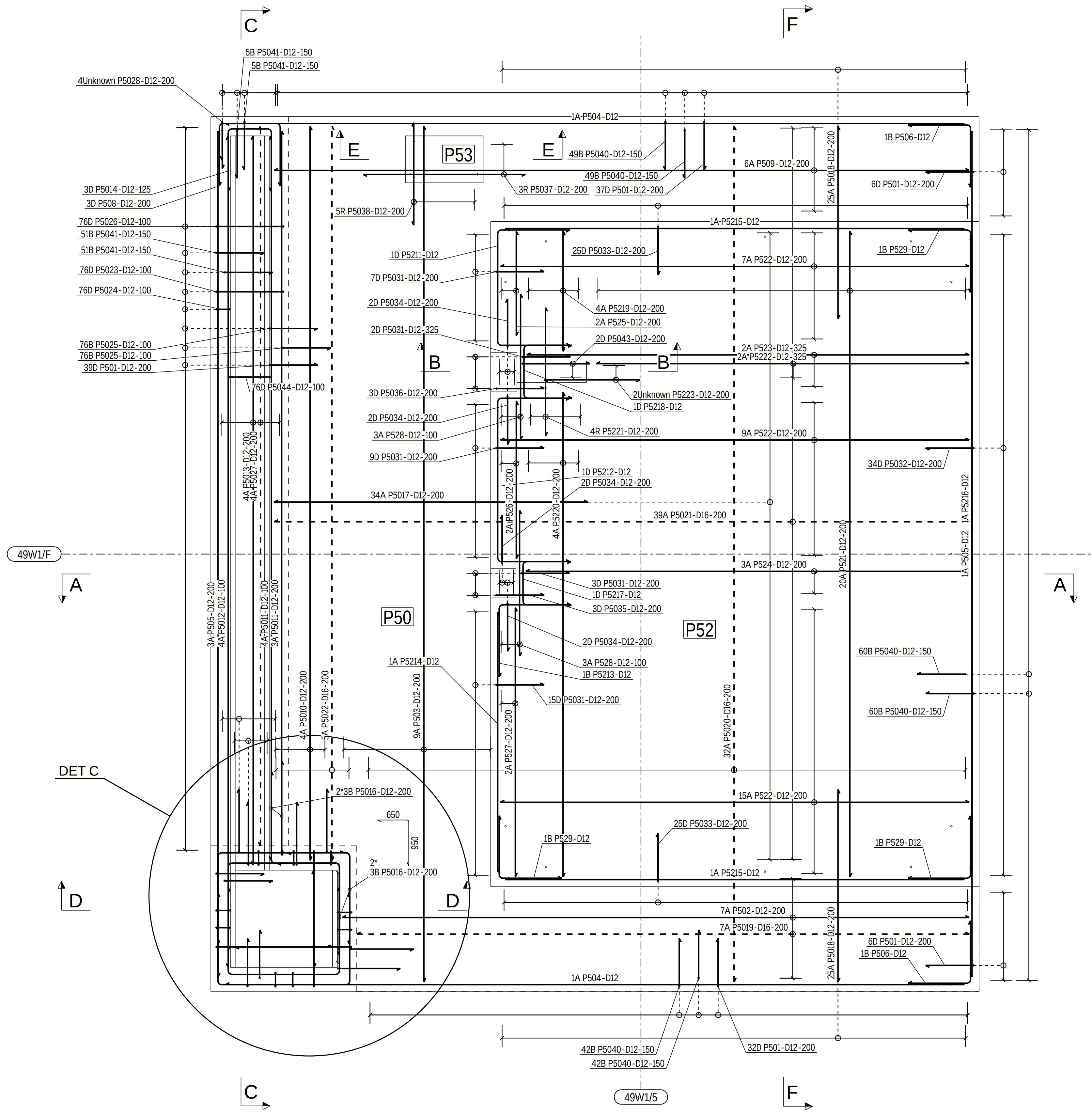
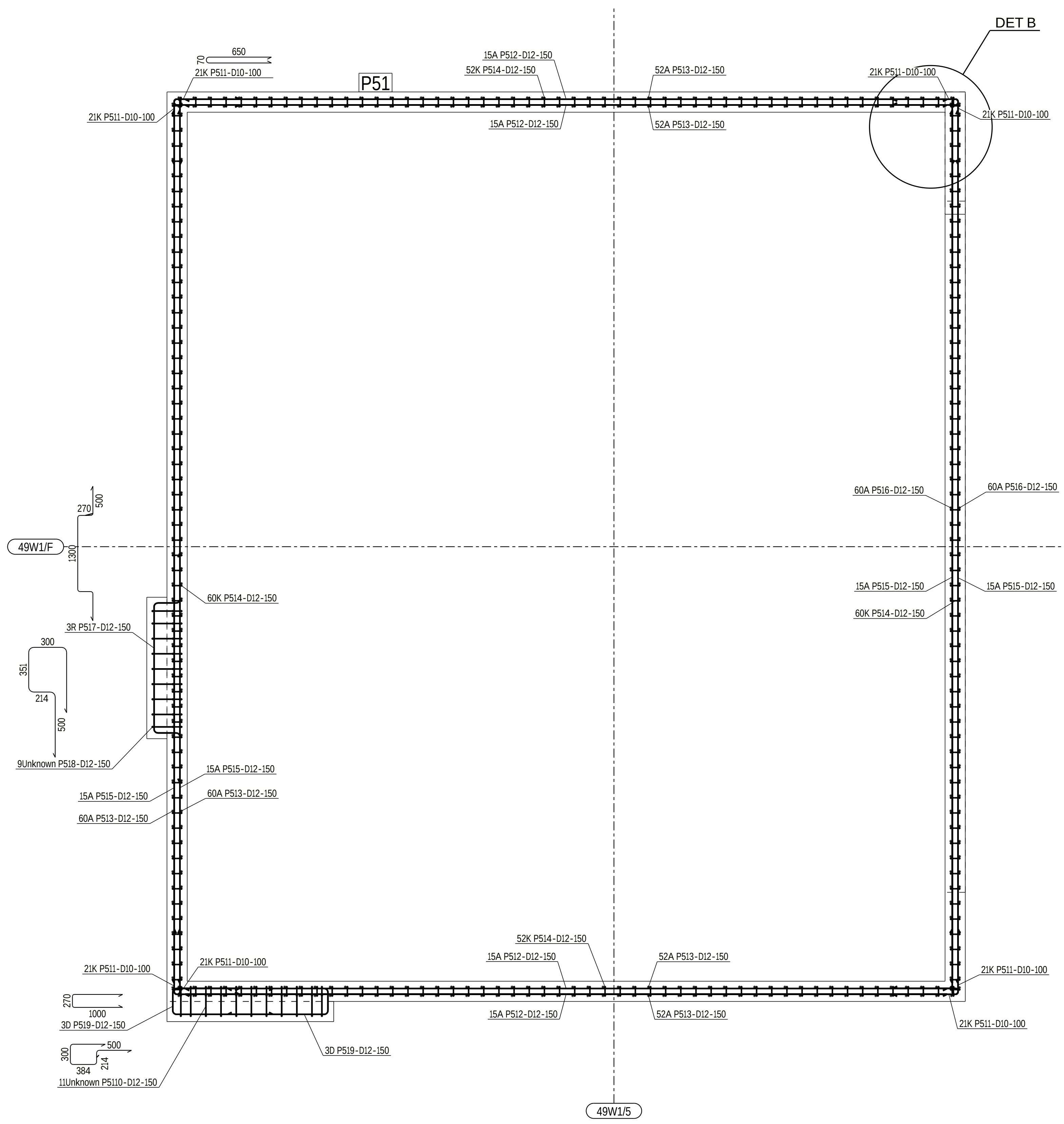
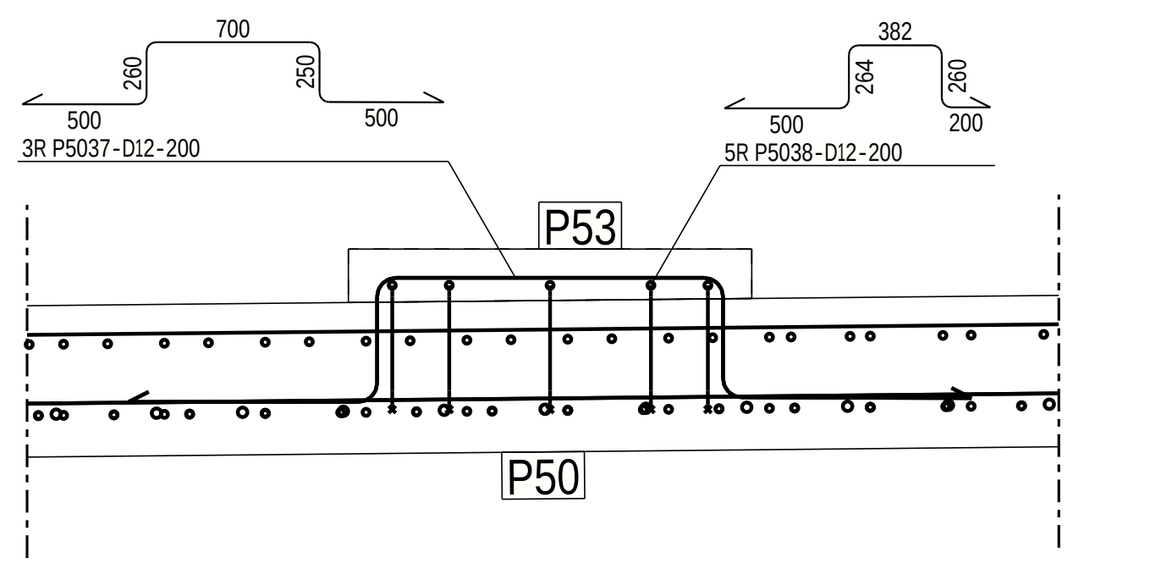
MD-216-49W1-EG-CV-DED-1020
DRAWING NUMBER
REV
D01



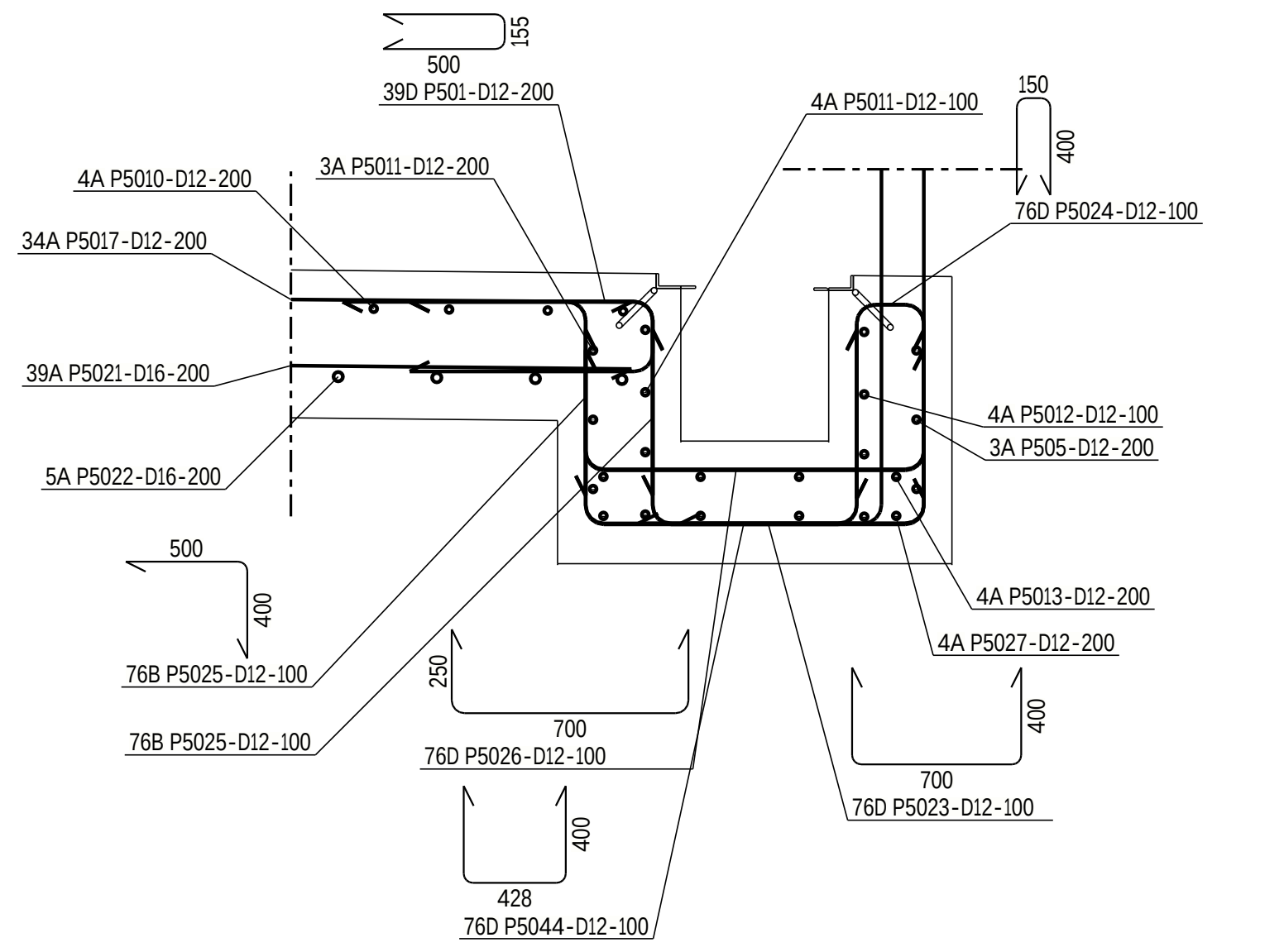
## R.C.C GROUND SLAB REINFORCEMENT
( SCALE 1:30 )



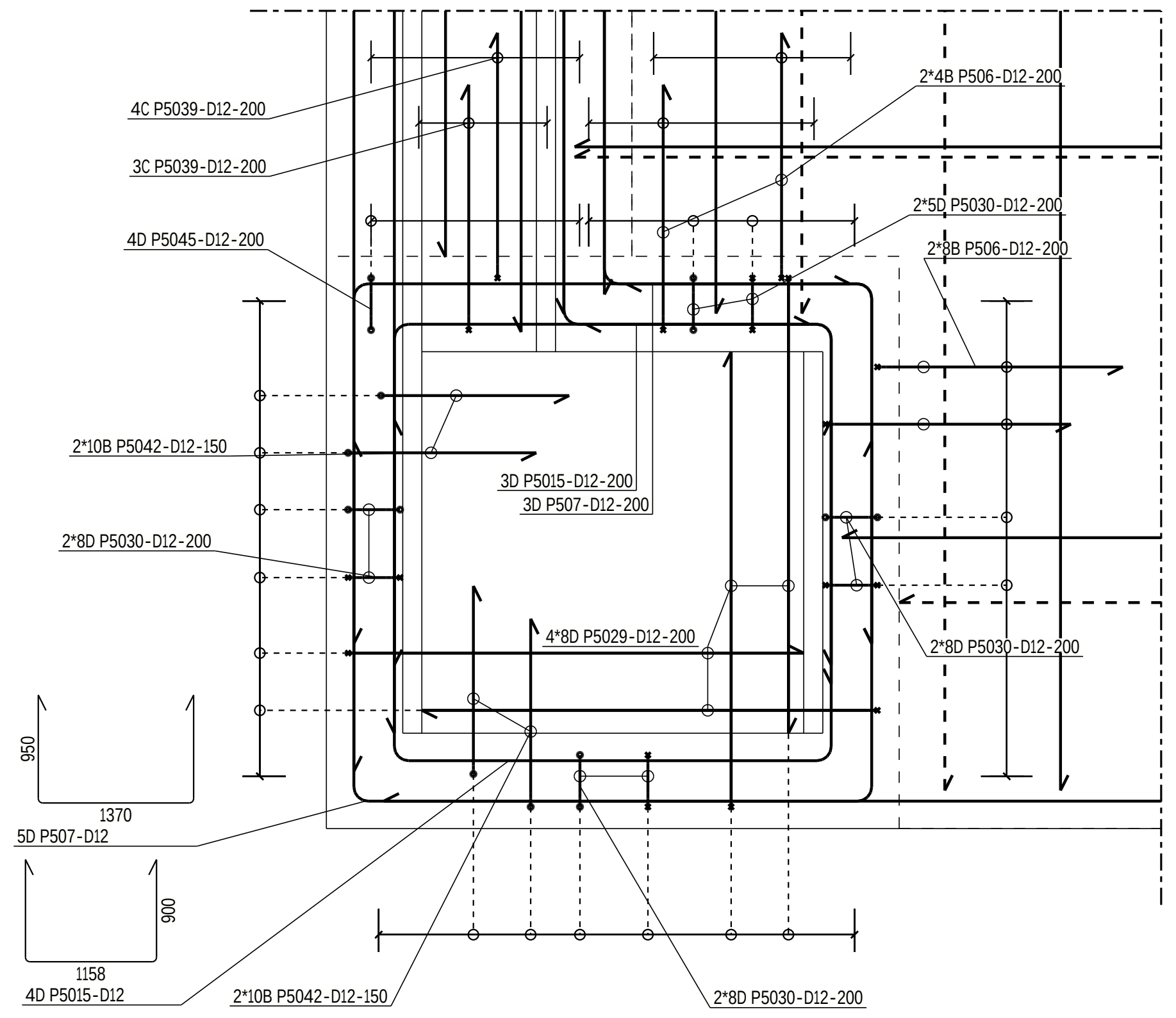
## R.C.C PIT WALL REINFORCEMENT
( SCALE 1:30 )



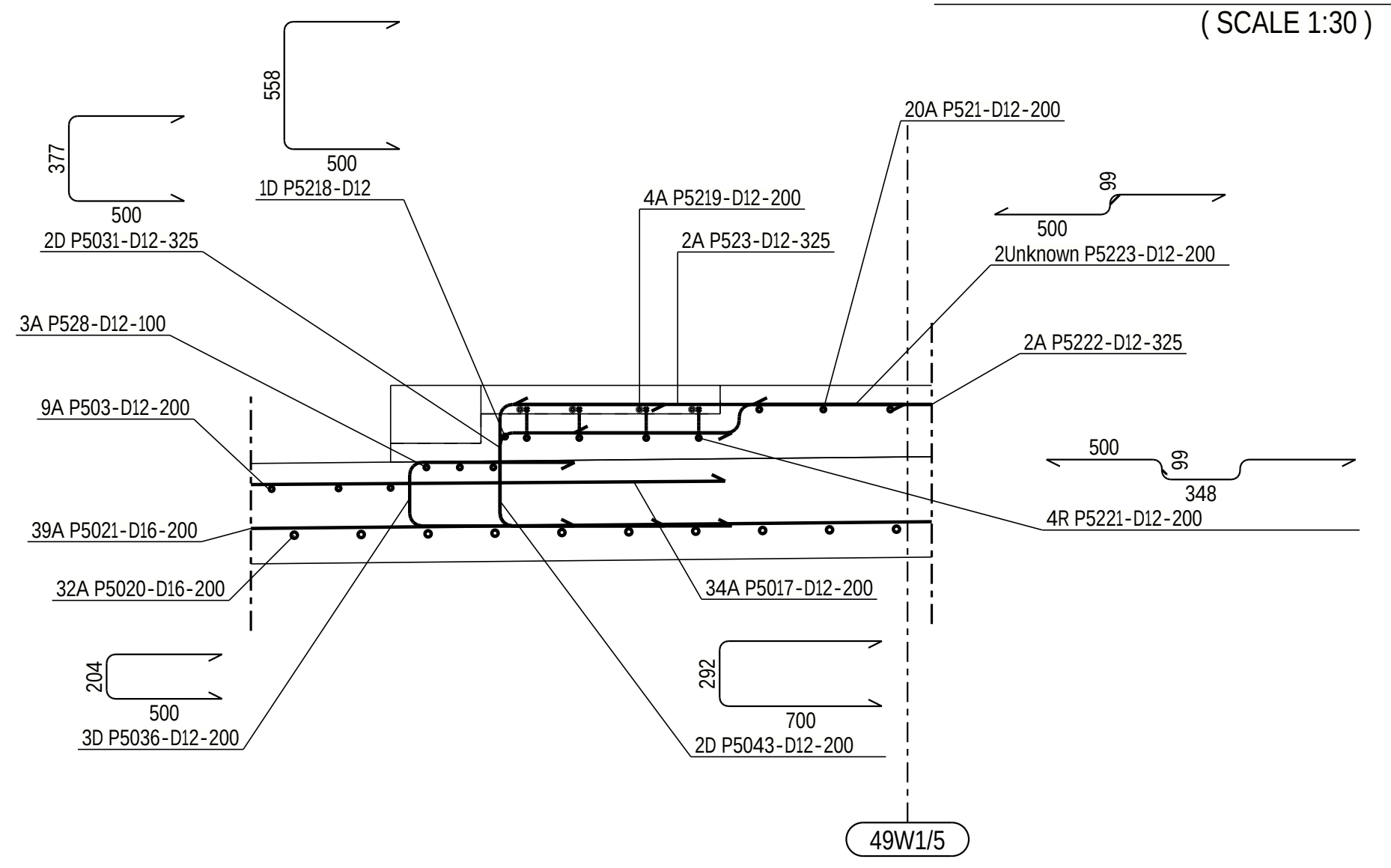
## SECTION : E - E
( SCALE 1:15 )



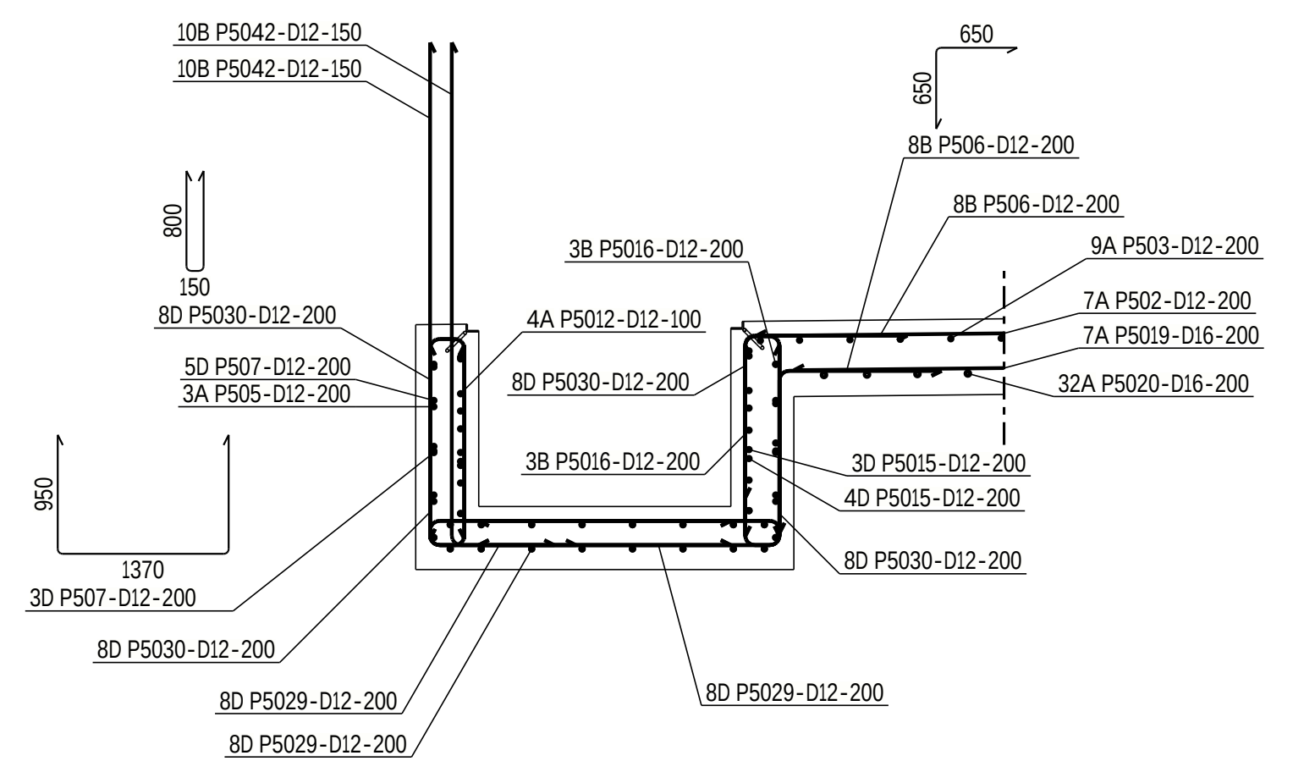
## DETAIL : A
( SCALE 1:15 )



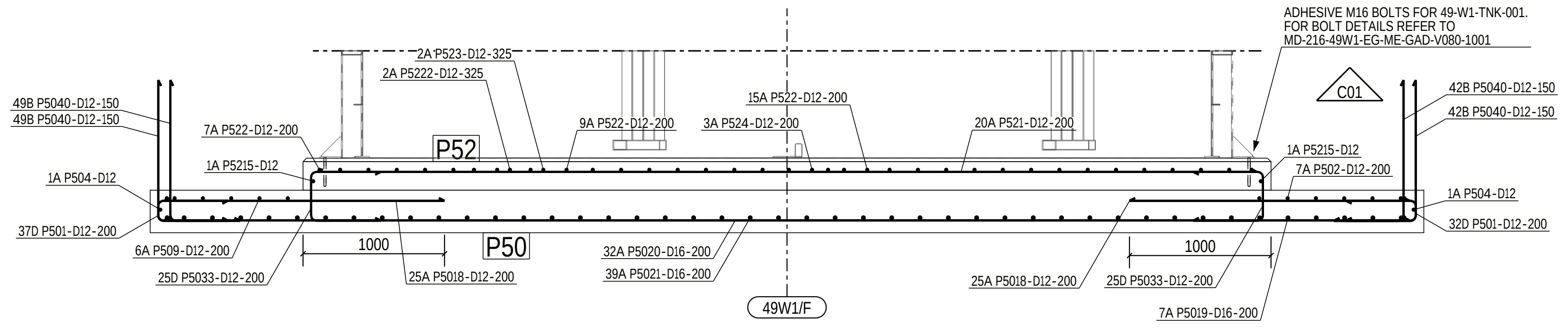
## DETAIL : C
( SCALE 1:15 )



## SECTION : B - B
( SCALE 1:20 )

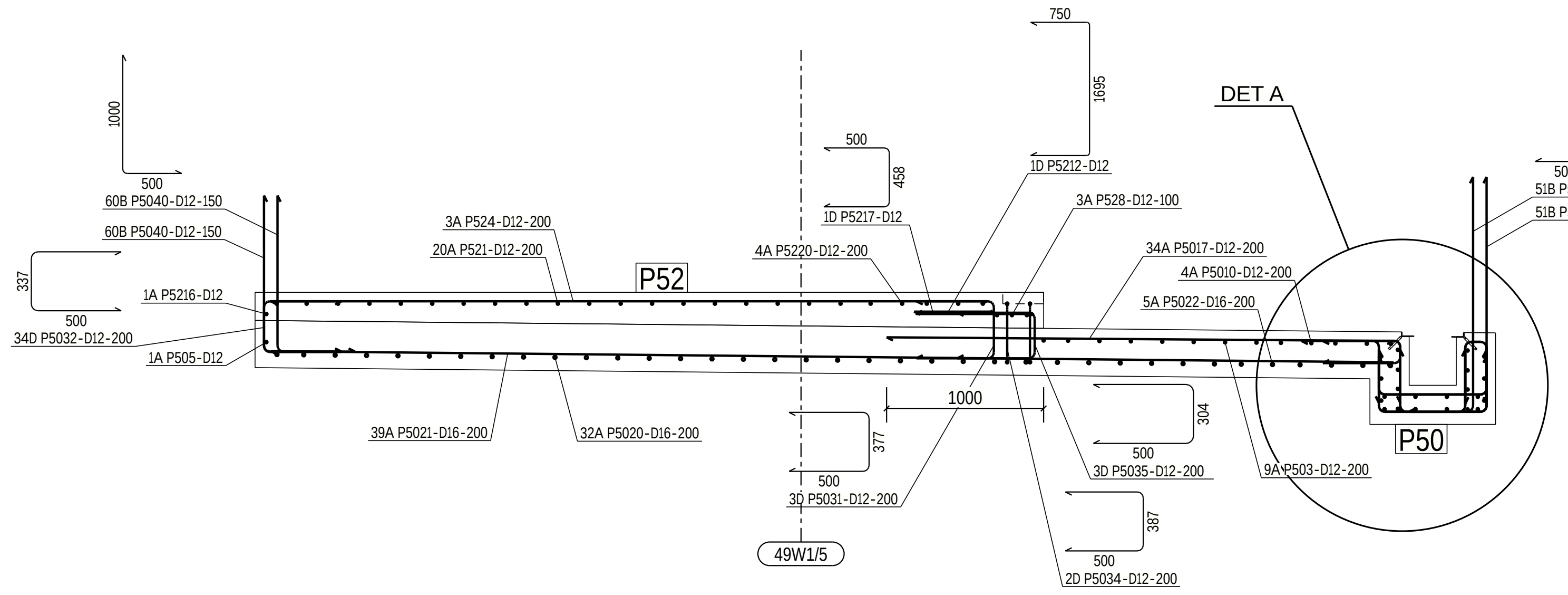


## SECTION : D - D
( SCALE 1:30 )

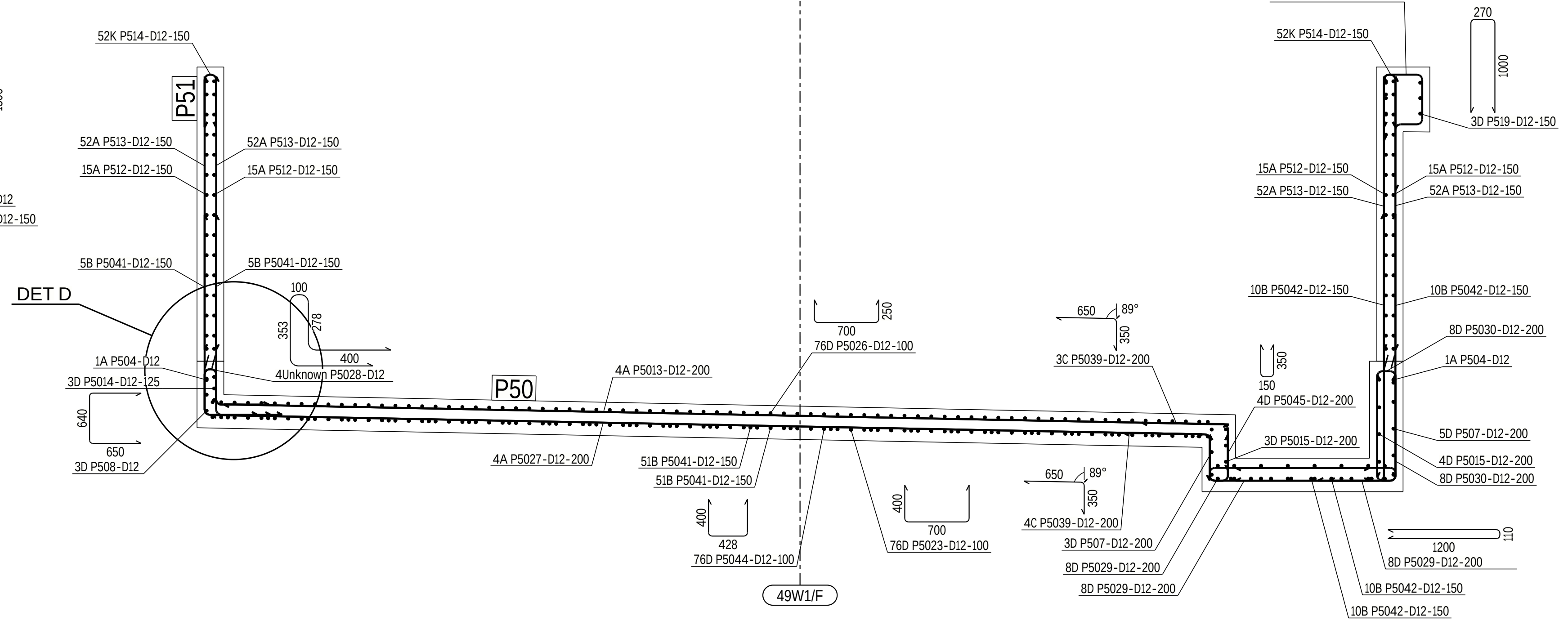


ADHESIVE M16 BOLTS FOR 49-W1-TNK-001.
FOR BOLT DETAILS REFER TO
MD-216-49W1-EG-ME-GAD-V080-1001
C01
P52
P50
49W1/F

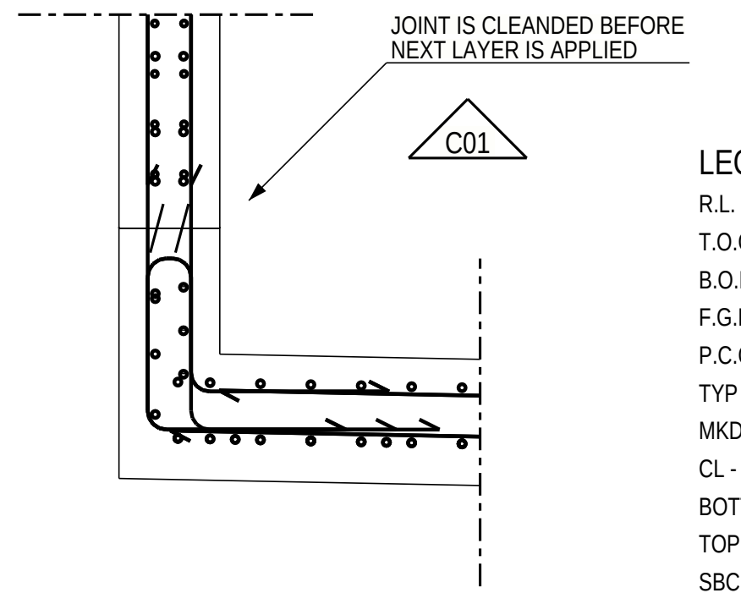
## SECTION : F - F
( SCALE 1:30 )



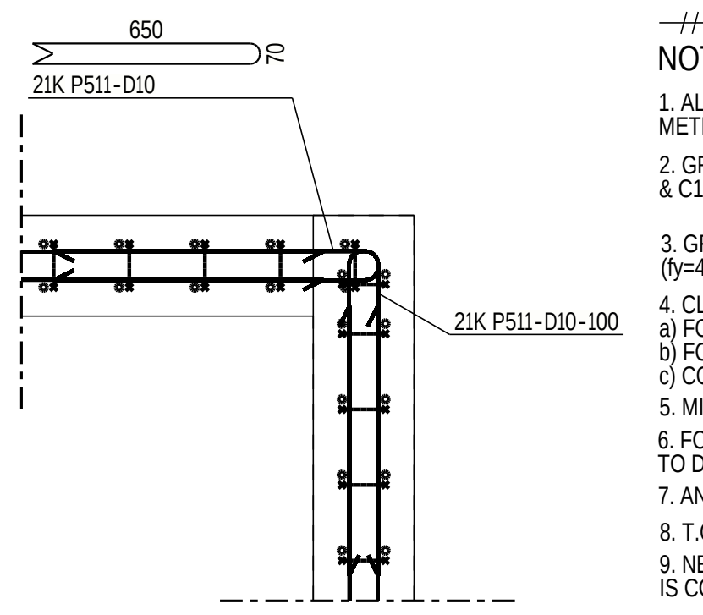
## SECTION : A - A
( SCALE 1:30 )



## SECTION : C - C
( SCALE 1:30 )



## DETAIL : D
## CONSTRUCTION JOINT DETAIL
( SCALE 1:15 )



## DETAIL : B
( SCALE 1:15 )

LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE
—//— - CONSTRUCTION JOINT

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

## CONCRETE INFORMATION

| | | |
|---|---|---|
| Concretes used | C35 | |
| Concrete cover 1 | 75±10mm | BOTTOM OF SLAB |
| Concrete cover 2 | 50±10mm | SIDES AND TOP OF SLAB, WALLS |

## REINFORCING BAR LIST
CAST UNIT CODE: GROUT PCS: 3

## REINFORCING BAR LIST
CAST UNIT CODE: P50 PCS: 1

| TYPE | POS | PCS | GRADE | D [mm] | L [mm] | dL [mm] | WEIGHT SUM [kg] | a | b | c | d | e | u | v | x | TD | COMMENT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D | P501 | 120 | B500B | 12 | 1090 | | 116.7 | 500 | 155 | 500 | | | | | | 60 | |
| A | P502 | 7 | B500B | 12 | 6450 | | 40.1 | 6450 | | | | | | | | | |
| A | P503 | 9 | B500B | 12 | 8800 | | 70.3 | 8800 | | | | | | | | | |
| A | P504 | 2 | B500B | 12 | 7600 | | 13.5 | 7600 | | | | | | | | | |
| A | P505 | 4 | B500B | 12 | 8700 | | 30.9 | 8700 | | | | | | | | | |
| B | P506 | 26 | B500B | 12 | 1270 | | 29.3 | 650 | 650 | | | | 90 | | | 60 | |
| D | P507 | 8 | B500B | 12 | 3210 | | 22.8 | 950 | 1370 | 950 | | | | | | 60 | |
| D | P508 | 3 | B500B | 12 | 1880 | | 5.0 | 650 | 640 | 650 | | | | | | 60 | |
| A | P509 | 6 | B500B | 12 | 7150 | | 38.1 | 7150 | | | | | | | | | |
| A | P5010 | 4 | B500B | 12 | 7550 | | 26.8 | 7550 | | | | | | | | | |
| A | P5011 | 7 | B500B | 12 | 7450 | | 46.3 | 7451 | | | | | | | | | |
| A | P5012 | 4 | B500B | 12 | 8550 | | 30.4 | 8551 | | | | | | | | | |
| A | P5013 | 4 | B500B | 12 | 7500 | | 26.6 | 7501 | | | | | | | | | |
| D | P5014 | 3 | B500B | 12 | 1690 | | 4.5 | 650 | 458 | 650 | | | | | | 60 | |
| D | P5015 | 7 | B500B | 12 | 2890 | | 18.0 | 900 | 1158 | 900 | | | | | | 60 | |
| B | P5016 | 12 | B500B | 12 | 1570 | | 16.7 | 950 | 650 | | | | 90 | | | 60 | |
| A | P5017 | 34 | B500B | 12 | 3220 | | 97.5 | 3228 | | | | | | | | | |
| A | P5018 | 50 | B500B | 12 | 1980 | | 87.9 | 1980 | | | | | | | | | |
| A | P5019 | 7 | B500B | 16 | 6300 | | 69.7 | 6300 | | | | | | | | | |
| A | P5020 | 32 | B500B | 16 | 8800 | | 444.9 | 8800 | | | | | | | | | |
| A | P5021 | 39 | B500B | 16 | 7150 | | 440.6 | 7150 | | | | | | | | | |
| A | P5022 | 5 | B500B | 16 | 7550 | | 59.6 | 7550 | | | | | | | | | |
| D | P5023 | 76 | B500B | 12 | 1440 | | 97.2 | 400 | 700 | 400 | | | | | | 60 | |
| D | P5024 | 76 | B500B | 12 | 890 | | 60.1 | 400 | 150 | 400 | | | | | | 60 | |
| B | P5025 | 152 | B500B | 12 | 870 | | 117.4 | 400 | 500 | | | | 90 | | | 60 | |
| D | P5026 | 76 | B500B | 12 | 1140 | | 77.0 | 250 | 700 | 250 | | | | | | 60 | |
| A | P5027 | 4 | B500B | 12 | 7400 | | 26.3 | 7401 | | | | | | | | | |
| ** | P5028 | 4 | B500B | 12 | 1410 | | 5.0 | | | | | | | | | | |
| D | P5029 | 32 | B500B | 12 | 2450 | | 69.6 | 1200 | 110 | 1200 | | | | | | 60 | |
| D | P5030 | 58 | B500B | 12 | 1690 | | 87.0 | 800 | 150 | 800 | | | | | | 60 | |
| D | P5031 | 36 | B500B | 12 | 1310 | | 42.1 | 500 | 377 | 500 | | | | | | 60 | |
| D | P5032 | 34 | B500B | 12 | 1270 | | 38.6 | 500 | 337 | 500 | | | | | | 60 | |
| D | P5033 | 50 | B500B | 12 | 1320 | | 57.9 | 500 | 385 | 500 | | | | | | 60 | |
| 39 | AT GROUP | 25 | PCS | ... | 1280 | 2 | ... | 500 | 340 | 500 | | | | | | | |
| D | P5034 | 8 | B500B | 12 | 1320 | | 9.4 | 500 | 387 | 500 | | | | | | 60 | |
| D | P5035 | 3 | B500B | 12 | 1240 | | 3.3 | 500 | 304 | 500 | | | | | | 60 | |
| D | P5036 | 3 | B500B | 12 | 1140 | | 3.0 | 500 | 204 | 500 | | | | | | 60 | |
| R | P5037 | 3 | B500B | 12 | 2090 | | 5.6 | 500 | 250 | 700 | 260 | 500 | | | | 60 | |
| R | P5038 | 5 | B500B | 12 | 1480 | | 6.6 | 500 | 264 | 382 | 260 | 200 | | | | 60 | |
| C | P5039 | 7 | B500B | 12 | 970 | | 6.0 | 650 | 350 | | | | 89 | | | 60 | |
| B | P5040 | 302 | B500B | 12 | 1470 | | 394.2 | 1000 | 500 | | | | 90 | | | 60 | |
| B | P5041 | 112 | B500B | 12 | 1970 | | 195.9 | 1500 | 500 | | | | 90 | | | 60 | |
| B | P5042 | 40 | B500B | 12 | 2470 | | 87.7 | 2000 | 500 | | | | 90 | | | 60 | |
| D | P5043 | 2 | B500B | 12 | 1630 | | 2.9 | 700 | 292 | 700 | | | | | | 60 | |
| D | P5044 | 76 | B500B | 12 | 1160 | | 78.8 | 400 | 428 | 400 | | | | | | 60 | |
| D | P5045 | 4 | B500B | 12 | 790 | | 2.8 | 350 | 150 | 350 | | | | | | 60 | |

REINFORCING BAR TOTAL WEIGHT [kg] 3211.1

## REINFORCING BAR LIST
CAST UNIT CODE: P51 PCS: 1

| TYPE | POS | PCS | GRADE | D [mm] | L [mm] | dL [mm] | WEIGHT SUM [kg] | a | b | c | d | e | u | v | x | TD | COMMENT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| K | P511 | 168 | B500B | 10 | 1320 | | 137.2 | 650 | 650 | | | | | | | 40 | |
| A | P512 | 60 | B500B | 12 | 7540 | | 401.7 | 7540 | | | | | | | | | |
| A | P513 | 328 | B500B | 12 | 2040 | | 594.2 | 2040 | | | | | | | | | |
| K | P514 | 224 | B500B | 12 | 840 | | 167.1 | 400 | 400 | | | | | | | 60 | |
| A | P515 | 60 | B500B | 12 | 8640 | | 460.3 | 8640 | | | | | | | | | |
| A | P516 | 120 | B500B | 12 | 1860 | | 198.3 | 1861 | | | | | | | | | |
| R | P517 | 3 | B500B | 12 | 2720 | | 7.2 | 500 | 270 | 1300 | 270 | 500 | | | | 60 | |
| ** | P518 | 9 | B500B | 12 | 1740 | | 14.0 | | | | | | | | | | |
| D | P519 | 6 | B500B | 12 | 2210 | | 11.8 | 1000 | 270 | 1000 | | | | | | 60 | |
| ** | P5110 | 11 | B500B | 12 | 1770 | | 17.4 | | | | | | | | | | |

REINFORCING BAR TOTAL WEIGHT [kg] 2009.2

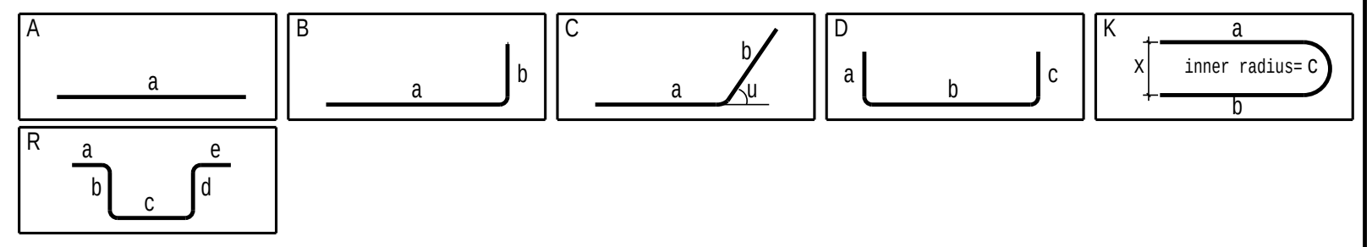
## REINFORCING BAR LIST
CAST UNIT CODE: P52 PCS: 1

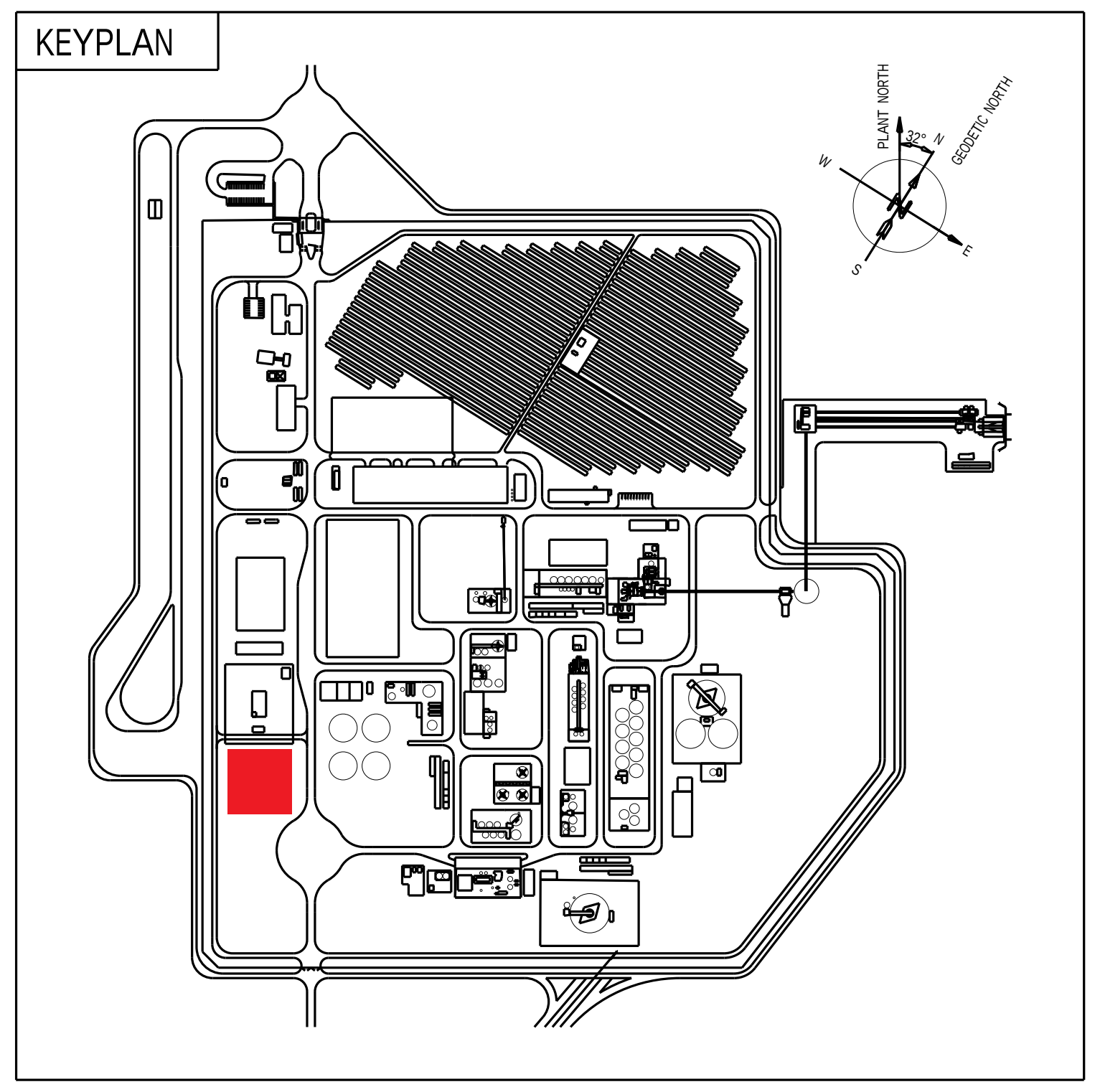
| TYPE | POS | PCS | GRADE | D [mm] | L [mm] | dL [mm] | WEIGHT SUM [kg] | a | b | c | d | e | u | v | x | TD | COMMENT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | P521 | 20 | B500B | 12 | 6640 | | 117.9 | 6640 | | | | | | | | | |
| A | P522 | 31 | B500B | 12 | 4820 | | 132.7 | 4822 | | | | | | | | | |
| A | P523 | 2 | B500B | 12 | 4550 | | 8.1 | 4552 | | | | | | | | | |
| A | P524 | 3 | B500B | 12 | 4560 | | 12.2 | 4562 | | | | | | | | | |
| A | P525 | 2 | B500B | 12 | 1070 | | 1.9 | 1075 | | | | | | | | | |
| A | P526 | 2 | B500B | 12 | 1620 | | 2.9 | 1625 | | | | | | | | | |
| A | P527 | 2 | B500B | 12 | 2770 | | 4.9 | 2770 | | | | | | | | | |
| A | P528 | 6 | B500B | 12 | 1500 | | 8.0 | 1500 | | | | | | | | | |
| B | P529 | 3 | B500B | 12 | 1270 | | 3.4 | 650 | 650 | | | | 90 | | | 60 | |
| D | P5211 | 1 | B500B | 12 | 2640 | | 2.3 | 750 | 1200 | 750 | | | | | | 60 | |
| D | P5212 | 1 | B500B | 12 | 3130 | | 2.8 | 750 | 1695 | 750 | | | | | | 60 | |
| B | P5213 | 1 | B500B | 12 | 1470 | | 1.3 | 750 | 750 | | | | 90 | | | 60 | |
| A | P5214 | 1 | B500B | 12 | 2670 | | 2.4 | 2670 | | | | | | | | | |
| A | P5215 | 2 | B500B | 12 | 4720 | | 8.4 | 4722 | | | | | | | | | |
| A | P5216 | 1 | B500B | 12 | 6540 | | 5.8 | 6540 | | | | | | | | | |
| D | P5217 | 1 | B500B | 12 | 1390 | | 1.2 | 500 | 458 | 500 | | | | | | 60 | |
| D | P5218 | 1 | B500B | 12 | 1490 | | 1.3 | 500 | 558 | 500 | | | | | | 60 | |
| A | P5219 | 4 | B500B | 12 | 1230 | | 4.4 | 1235 | | | | | | | | | |
| A | P5220 | 4 | B500B | 12 | 4980 | | 17.7 | 4985 | | | | | | | | | |
| R | P5221 | 4 | B500B | 12 | 1420 | | 5.1 | 500 | 99 | 348 | 99 | 500 | | | | 60 | |
| A | P5222 | 2 | B500B | 12 | 3830 | | 6.8 | 3837 | | | | | | | | | |
| ** | P5223 | 2 | B500B | 12 | 1030 | | 1.8 | | | | | | | | | | |

REINFORCING BAR TOTAL WEIGHT [kg] 353.4

## REINFORCING BAR LIST
CAST UNIT CODE: P53 PCS: 1

REINFORCEMENT BENDING SHAPES:





| DOCUMENT NUMBER | REFERENCE DOCUMENTS | DOCUMENT NUMBER | REFERENCE DOCUMENTS |
|---|---|---|---|
| MD-216-49W1-EG-CV-DED-1011 | 49W1-TNK-001 PIT DIMENSIONAL DRAWING | | |
| MD-216-49W1-EG-ME-GAD-V080-1001 | 49W1-TNK-001, Coagulant Mixin Basin | MD-216-49W1-EG-CV-CAL-1006 | DAF BASIN, COAGULANT BASIN, PIT, SLUDGE BASIN, VESSEL FOUNDATION |
| MD-216-49W1-EG-CV-GAD-1003 | 1-10/A-E GENERAL ARRANGEMENT DRAWING | MD-216-1000-EG-ST-CRT-1001 | DESIGN CRITERIA |
| MD-216-49W1-EG-CV-GAD-1001 | 1-8/E-J GENERAL ARRANGEMENT DRAWING | MD-216-49W1-EG-CV-GAD-1002 | 7-12/D-J GENERAL ARRANGEMENT DRAWING |

| REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|
| D01 | 29.01.2024 | AS-BUILT | JonRen | MauNot | LuiPiz | SouCha | |
| C01 | 12.01.2022 | ISSUED FOR CONSTRUCTION | JonRen | MauNot | NanWu | LuiPiz | |
| A01 | 17.12.2021 | ISSUED FOR APPROVAL | JonRen | MauNot | NanWu | LuiPiz | |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to recieve it or be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

| | | |
|---|---|---|
| LARSEN & TOUBRO DRAWING NUMBER | O19045_080_C_D_24892_020 | REVISION 2 |
| OUTOTEC DRAWING NUMBER | 900707_GAU01_AG11_CTB01_49020 | DOCUMENT ID OU100736746 — REVISION 2 |
| DRAWN | JonRen | 17.12.2021 |
| CHECKED | MauNot | 17.12.2021 |
| ENGINEER | NanWu | 17.12.2021 |
| APPROVED | LuiPiz | 17.12.2021 |
| EMPLOYER | | |
| SCALE | | A0 |

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
49W1-TNK-001 PIT
REINFORCEMENT DRAWING

DRAWING NUMBER
MD-216-49W1-EG-CV-DED-1020
REV
D01